6G RESEARCH

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Introduction: Global Transition from 5G-Advanced to 6G

The global telecommunications ecosystem is on the threshold of a fundamental structural paradigm shift. As fifth-generation (5G) networks and their normative evolution, 5G-Advanced, reach their commercial maturity phase, the industry faces the inescapable reality that the current architecture has physical, spectral, and topological limits that restrict its capacity to support the demands of the next decade.[1] The cyber ecosystem projected for 2030 requires an absolute convergence among the physical, digital, and biological spheres, driven by ubiquitous artificial intelligence (AI), collaborative robotics, and the massive creation of digital twins.[1] These projected use cases demand network capabilities that far exceed the theoretical limits of 5G's radio frequency (RF) and core network infrastructures.[2]

This research report details a comprehensive and detailed analysis of the current state of mobile telecommunications evolution. The document is divided into two main analytical blocks. **Block 1** dissects the "exhaustion of 5G," evaluating the technical limitations documented in academic research from the IEEE Xplore repository, including bottlenecks in spectral capacity, critical asymmetries in the uplink due to generative artificial intelligence, and thermodynamic barriers in energy efficiency.[2] **Block 2** maps the official timeline and the global standardization architecture, critically analyzing the International Telecommunication Union (ITU) IMT-2030 guiding framework [5], the release cadence of the 3rd Generation Partnership Project (3GPP) [7], and the divergent socio-technical stances among regional consortia, specifically the "National 6G Roadmap" by the Next G Alliance in North America [9] and the vision and societal values deliverables by the Hexa-X consortium in the European Union.[10]

Block 1: Technical Exhaustion and Limitations of 5G and 5G-Advanced

The development of 5G-Advanced, formally initiated with 3GPP Release 18, was conceived as an evolutionary bridge aimed at extracting the maximum possible efficiency from the original 5G network architecture.[2] Although it introduced substantial improvements in the management of reduced capability (RedCap) devices, multiple-input multiple-output (MIMO) performance, and initial support for extended reality (XR), technical literature and deployment trials have exposed critical vectors of "exhaustion".[2] This exhaustion does not imply a network failure, but rather approaching the theoretical frontiers dictated by the laws of wave physics and the foundational architectural decisions of the previous decade.

1.1. Spectral Saturation and the Limits of Shannon Capacity

The primary catalyst justifying the imminent need for 6G is the severe congestion and spectral saturation in bands below 7 GHz (sub-7 GHz).[2] Current 5G deployments rely heavily on mid-band spectrum, particularly frequencies around 3.5 GHz, as they offer an optimal physical balance between signal propagation for wide-area coverage and the bandwidth needed to support high-throughput capabilities.[2] However, as subscriber density and device performance requirements exponentially increase, the usable space in these bands is rapidly depleting, leaving operators without the contiguous spectral resources necessary to expand the network.[2]

From an information theory perspective, the capacity of a wireless communication channel is inexorably governed by the Shannon-Hartley theorem. The maximum theoretical capacity is defined by the equation $C = B\log_2(1 + S/N)$, where **C** represents the channel capacity, **B** is the available bandwidth, **S** is the transmitted signal power, and **N** is the underlying noise power. In the 5G-Advanced framework, the industry has pushed modulation techniques almost to their Shannon limit through schemes like 256-QAM and research aimed at 1024-QAM. Concurrently, the signal power variable (**S**) cannot be indefinitely increased due to strict international regulations on electromagnetic emissions and the thermal and battery constraints of user equipment (UE).[2]

Consequently, the only mathematically viable vector to achieve peak data rates exceeding 100 gigabits per second (Gbps) or even approaching one terabit per second (Tbps) is a drastic and unprecedented expansion of bandwidth (**B**).[2] To navigate around this exhaustion, 6G-oriented research is aggressively steering towards two new spectral frontiers, each with enormous propagation challenges:

The Centimeter Range (7–24 GHz)

Identified through the Next G Alliance priorities and 3GPP channeling studies in Release 19, the centimeter-wave spectrum is considered crucial for the early commercial implementation phase of 6G.[8] This range provides contiguous spectrum blocks significantly wider than the sub-7 GHz bands, maintaining somewhat more manageable propagation characteristics compared to millimeter waves, enabling an essential balance between continuous coverage and dense capacity.[2]

The Sub-Terahertz (Sub-THz) Frontier (>100 GHz)

Considered the ultimate enabler of extreme-speed services, the sub-THz band offers ultra-wide signal bandwidths. Hexa-X architectures and IEEE analyses indicate that exploiting frequencies above 100 GHz is mandatory for "extreme experience" scenarios, such as wireless data centers, ultra-dense hotspot offloading, and cellular wireless backhaul.[1] However, migrating to this band presents massive technological exhaustion barriers for current 5G transceivers, including the imperative need for channel modeling that accounts for severe atmospheric molecular absorption, the design of highly energy-efficient ultra-dense antenna arrays, and nearly microscopic precision beam tracking to overcome the extreme path loss inherent to these frequencies.[2]

Spectral Band	Status in 5G-Advanced	6G Architectural Projection	Underlying Research Challenge
Sub-7 GHz	Highly saturated; current main workhorse.[2]	Continuous use through Multi-RAT Spectrum Sharing (MRSS) with 5G.[7]	Maximizing spatial reuse without interference.
7–24 GHz (Centimeter)	Limited or no normative deployment.	"Day 1" priority spectrum for initial deployments and area	Robust channel modeling and indoor structure penetration.[8]

		capacity.[8]	
Sub-THz (>100 GHz)	Beyond the scope of the current standard.	Very high-capacity fixed links, terabit offloading, fronthaul fusion.[2]	High path loss, molecular absorption, Extreme MIMO requirements.[2]

1.2. The Agentic AI Crisis and the Uplink Bottleneck

Historically, the foundational architecture of cellular mobile networks, from third-generation (3G) networks to the current 5G, has been designed based on an asymmetrical traffic paradigm heavily skewed towards data downloading (downlink).[4] The configurations of time slots in Time Division Duplex (TDD) in 5G are optimized assuming that users consume much more information from the network, such as high-definition video streaming, compared to the amount of data they send to the network.[4]

The imminent maturation and ubiquity of Generative AI (GenAI) and, more specifically, "Agentic AI," fundamentally shatters this architectural assumption.[3] Agentic AI refers to artificially intelligent systems and devices that act as autonomous agents on behalf of the user, making real-time decisions and reacting to environmental stimuli.[4] Future edge devices, such as extended reality-enabled smart glasses, brain-computer interfaces, and embodied agents in manufacturing, will require sending continuous and massive streams of high-fidelity telemetry, 3D spatial maps (point clouds), uncompressed video, and sensor data directly to the network edge cloud for processing and execution of inference models.[3]

According to corporate and standardization analytical reports, this massive and constant flow will irreversibly transform the uplink into the main "bottleneck" of mobile telecommunications.[4] 5G-Advanced lacks the flexibility and raw capacity to dynamically resize TDD divisions to the scale required by Agentic AI without severely degrading downlink performance for the rest of the cell. This level of exhaustion forces 6G design to investigate completely disruptive technologies, most notably Full-Duplex communication operations (simultaneous transmission and reception over the same frequency band, canceling self-interference) and the native integration of AI-based channel coding that reduces user transmission overheads by efficiently packaging the semantic state of the data rather than simply transporting raw bits.[2]

1.3. Energy Efficiency, Diminishing Returns, and Integral Sustainability

Thermodynamic exhaustion is one of the most limiting vectors of the 5G ecosystem. Despite regulatory efforts, approximately 70% of a telecommunications operator's total energy consumption is directly concentrated in the radio access network (RAN) base station.[2] To contain this energy hemorrhage, 5G-Advanced introduced and refined advanced power-saving or "sleep" modes, allowing the base station to dynamically deactivate transceiver chains during microseconds or hours of low traffic demand.[2]

However, research published in IEEE Xplore shows that current energy mitigation solutions are hitting a wall of diminishing returns.[2] Further reducing base station transmission activity creates unsustainable negative externalities: if the base station remains inactive for prolonged periods, it becomes mathematically and operationally too difficult for user equipment (UE) to discover synchronization signals, perform accurate channel measurements, and reconnect in a timely manner, which deteriorates the user experience and increases the mobile terminal's own battery consumption by forcing it to "scan" the environment for prolonged periods.[2]

For 6G, the paradigm must transmute from a mere search for operational "energy efficiency" to a framework of holistic sustainability from day zero. The Hexa-X-II consortium addresses this exhaustion by rethinking infrastructure deployment through the dichotomy of the "Sustainability Footprint" and the "Sustainability Handprint" [10]:

- **The Footprint:** Encompasses the first-order negative effects derived from the technology, such as the massive direct environmental impact of manufacturing, implementing, and powering millions of new 6G antennas and edge servers, as well as higher-order negative socioeconomic consequences.[10]
- **The Handprint:** Comprises the systemic positive effects that 6G-enabled solutions generate, allowing other industries (precision agriculture, autonomous traffic routing optimization, manufacturing automation) to massively reduce their own carbon and ecological footprints using connected intelligence, counteracting the network's initial environmental cost.[10]

To resolve the limitations of 5G sleep modes, 6G architectures propose AI-driven algorithmic orchestrations that predict cell load to intelligently modulate hardware, and fundamental research into zero-energy devices that do not require conventional batteries, harvesting energy through the environment's own radio frequency (RF) emissions or the sun, a key requirement for the massive densification of the Internet of Things (IoT).[10]

1.4. The Positioning Deficiency and the Demand for Integrated Sensing and Communication (ISAC)

The evolution of telecommunication services towards hyper-realistic digitization demands capabilities that go far beyond data transfer. In the 5G-Advanced paradigm, positioning techniques were improved to achieve accuracies in sub-metric ranges. While this proved adequate for general logistics and asset tracking, performance has maxed out against next-generation requirements.[2]

Foundational 6G applications, such as seamless immersive telepresence, the orchestration of cooperative robots (cobots) in agile factories, and the creation of massive Digital Twins at city or factory scale, inherently require spatial resolution and centimeter-level positioning, coupled with extremely low computational latencies of just a few tens of milliseconds.[2] Current 5G architectures treat positioning and communication as overlapping processes or auxiliary sequences (overlays), resulting in inefficient overheads.[2]

To bridge this capabilities gap, the 6G specification proposes total fusion at the physical layer (hardware and waveform) through Integrated Sensing and Communication (ISAC), also referred to as Joint Communication and Sensing (JCAS).[2] By leveraging the colossal bandwidths of the centimeter and sub-THz spectrums, alongside reconfigurable intelligent surfaces (RIS) and massive array antennas (Extreme MIMO), the cellular network literally transforms into a highly distributed bistatic or multistatic radar.[2] This native integration allows the same radio waveform to transfer data to a mobile terminal while simultaneously "bouncing" off physical objects in the environment, passively detecting the shape of buildings, tracking the flight of unmanned aerial vehicles (drones) to prevent collisions, and even monitoring rainfall or human biometric metrics without requiring the objects to possess a cellular transceiver, providing the digital twin with real-time updates.[2]

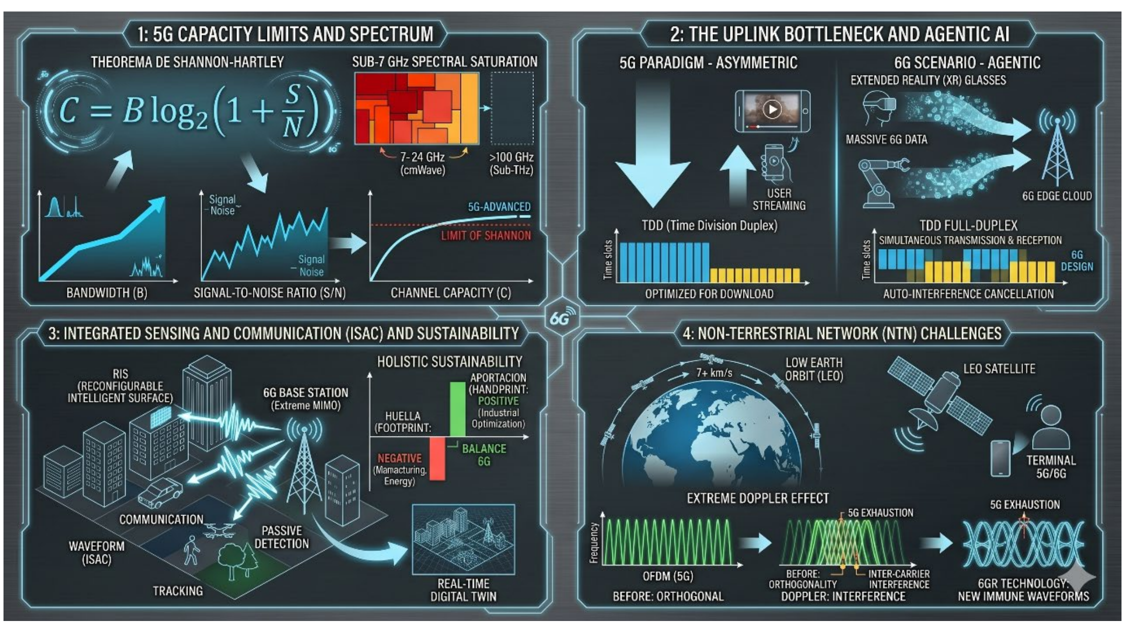
1.5. Topological Challenges: Non-Terrestrial Networks (NTN) and Core Decentralization

Finally, 5G network architecture has reached significant operational ceilings in their centralized switching topology and in the deployment of three-dimensional networks.

On one hand, the current cloud computing model associated with 5G typically consolidates data storage and processing resources in a small number of regional or national data centers, managing mobility through Centralized Mobility Management (CMM) systems.[9] This hyper-centralization imposes inexorable delays due to backhaul optical fiber propagation latency. For real-time haptic and cybernetic applications, this creates an insurmountable physical limit. Researchers propose Distributed Mobility Management (DMM) architectures to bring network anchors directly to the device's computational edge [18], aligning with the 6G vision of a "distributed cloud framework," where resources are osmotically dispersed in local edge servers, smart streetlights, and even the interconnected terminals themselves.[9]

On the other hand, the pursuit of true digital inclusion and uninterrupted global coverage requires the full integration of satellite infrastructures through Non-Terrestrial Networks (NTN).[1] 5G-Advanced initiated NTN protocols but has faced almost formidable astrodynamical obstacles. Low Earth Orbit (LEO) satellites move at relative speeds of over 7 kilometers per second relative to user equipment on the Earth's surface. This orbital velocity induces extreme Doppler frequency shifts.[2]

In 5G's underlying technology (Orthogonal Frequency-Division Multiplexing - OFDM), the massive Doppler shift breaks the essential orthogonality of the radio frequency subcarriers, causing devastating inter-carrier interference that destroys the transmitted information.[2] Furthermore, the timing advance synchronization required to compensate for massive space-to-ground propagation delays places immense



strain on current 5G protocols.[2] This barrier underpins the need for a completely autonomous 6G radio access technology (6GR), implementing new waveforms intrinsically immune or resilient to severe Doppler shifts and variable latencies to definitively merge terrestrial and space networks.[16]

Block 2: Official Timeline, Guiding Framework, and Global Standardization Architecture

The journey towards the commercial deployment of 6G, consensually slated for the 2030 horizon, is not a spontaneous leap, but rather a highly structured and legislated interconnected process by international organizations, corporate consortia, and strategic geopolitical alliances. This iterative process prevents the massive technical fragmentation that plagued earlier generations and ensures a standardized and economically viable hardware and software ecosystem on a global scale.[7]

2.1. ITU-R Recommendation M.2160 (IMT-2030 Framework): The Global Master Blueprint

The foundational scaffolding of the entire 6G ecosystem is dictated by the International Telecommunication Union (ITU). In November 2023, the ITU Radiocommunication Sector (ITU-R) published its official and guiding normative document: the *"Recommendation ITU-R M.2160-0: Framework and overall objectives of the future development of IMT for 2030 and beyond"*.[5] This framework establishes the imperative justification for the transgenerational transition and defines the ultimate objectives that any proposed technology must achieve to be certified as IMT-2030.

The central element of Recommendation M.2160 is the definition of capacity expectations through the so-called "palette diagram," which articulates fifteen critical capabilities, vastly surpassing the requirements of its predecessor IMT-2020 [20]:

- **Nine Evolved Capabilities:** Grounded in the pre-existing 5G metrics, such as extreme peak data rates, massive device densification, ultra-low latency, and improved mobility.[20] The numerical targets for these values push the thresholds of feasibility, forcing a rethinking of spectral bands.
- **Six Novel Capabilities:** This category enshrines the disruptive 6G paradigm, introducing unprecedented operational demands, most notably the native inclusion and integration of environmental sensing capacity and ubiquitous, integrated artificial intelligence (AI/ML) across all system layers.[5]

An underlying and vital principle in the document is that the ITU dictates that all target capacity values hold exactly the same priority for research and investigation purposes, meaning that achieving terabits per second rates cannot be sacrificed at the cost of abandoning the guiding principles of environmental sustainability, deep security, structural resilience, and inclusion ("connecting the unconnected" through affordable NTNs).[5]

2.2. The 3GPP Normative Sequence: From Release 18 to Release 21

The materialization of the ITU's theoretical vision rests on the shoulders of the 3rd Generation Partnership Project (3GPP). The 3GPP divides its activity into rigorously timed cycles called "Releases." The transition from 5G exhaustion to the 6G architecture spans the efforts specified from Release 18 up to the climax in Release 21.[7]

Maturation Limits: Release 18 and Release 19 (5G-Advanced)

Release 18, whose core specification concluded at the end of 2023, officially constituted the inauguration of the 5G-Advanced commercial era.[2] It fundamentally focused on stretching capacity boundaries through the continuous evolution of the MIMO system, the expansion of low-complexity RedCap terminals to scale industrial IoT, and the introduction of preliminary AI functions to improve network consumption efficiency.[2] It did not include native 6G studies.

Release 19, currently in normative progress with functional freeze and main conclusion projected for December 2025, fulfills an essential dual role.[8] On the one hand, it pragmatically addresses unresolved commercial needs of current operators. It introduces, for example, the extension of the *Layer 1/Layer 2 Triggered Mobility* (LTM) mechanism to allow seamless handovers between different gNBs (base stations), drastically reducing interruptions in latency-critical applications.[8] On the other hand, it serves as the prologue to 6G research: it authorizes two critical channel modeling studies that will pave the upcoming work, specifically analyzing radio wave response in the newly claimed 7 to 24 GHz spectrum and establishing the official channel model for future Integrated Sensing and Communication (ISAC) functionality.[8]

The Bridge to the Future: Release 20 (Start of 6G)

Release 20 represents the structural turning point of the entire mobile industry. This cycle, operationally scheduled to formally initiate and run between mid-2025 and conclude around June 2027 (stage freezes), is characterized by having a bifurcated scope: it will complete the final efforts of the 5G-Advanced deployment normative, while simultaneously launching the formal technical studies that will serve as the foundation for the 6G standard.[7]

In this Release, technical groups, including the Service and System Aspects (SA1) branch, agreed to allocate their times equally to study preliminary 6G requirements.[25] The study justifications outline a pragmatic philosophy, driven heavily by corporate lessons from the 2020s. During the development of 5G, the vast multiplicity of architectural options (complex transitions from Non-Standalone to Standalone networks) generated harmful market fragmentation that slowed operator deployments and limited agile monetization.[7]

Therefore, the central study justification for Release 20, termed "6G Radio" (6GR), agrees that the 6G architecture will adopt a Standalone (SA) approach from "Day 1," prioritizing an ultra-simplified system design and minimizing the adoption of multiple paralyzing options for the same operational functions.[17] This radio access technology (6GR) will be free from the burden of maintaining strict backward technical compatibility with the 5G NR waveform, allowing it to make disruptive performance leaps against the Doppler effect and spatial absorption.[16] Nevertheless, as a cost-mitigating strategy, the 6G RAN will be designed to interconnect in its initial stage to a smooth evolution of the existing 5G core (5GC), avoiding forcing operators to replace their entire network core simultaneously.[7]

The Final Normative Specification: Release 21

Once the exhaustive 21-month analytical cycles provided in Release 20 conclude, the 3GPP consortium will move from study to formal normative production in **Release 21**. The conclusive dates regarding the timeline borders for Release 21 will be subject to final agreement no later than the plenary meeting projected for June 2026.[7]

Release 21 will provide the first codified and binding set of technical specifications for manufacturing companies to begin molding 6G silicon, antennas, and switching equipment. Manufacturers like Ericsson postulate and trust that the working set of this Release 21 must be finalized and obtain a single drop code freeze by the end of 2028 (with alternative dates pointing to ASN.1 freezing around March 2029).[7] This margin is fundamental, given that it provides the mandatory window required for the 3GPP to submit its specifications through a formal self-evaluation process to the ITU in 2029, ensuring the standard's certification under the IMT-2030 designation, and facilitating the physical and commercial market release of the first globally operational 6G systems for the target year of 2030.[7]

Entity	**Cycle/Operational Phase**	**Estimated Timeframes**	**Deliverables and Specification Objectives**
3GPP	**Release 19**	Jan 2024 - Dec 2025	Inter-gNB mobility (LTM), Preliminary channel modeling for 7-24 GHz and ISAC.
3GPP	**Release 20**	Mid 2025 - Jun 2027	Final 5G-A split and 21 months of 6G technical study (6GR definition).
3GPP	**Release 21**	2027 - Dec 2028/2029	Generation of the first 6G technical norm in a single code drop.
ITU	**IMT-2030**	2028 - 2030	Reception of self-evaluations, ratification, and adoption of the official global standard.

2.3. Sociotechnical Priorities and Regional Axis Roadmaps

Simultaneously as standardizing institutions decide the physical parameters, the vision and justification for "why" develop these systems come from corporate-academic regional consortia. These consortia outline political and technological roadmaps to guarantee hegemony and geopolitical leadership in the impending 6G market.

North America: The Next G Alliance Strategy and the National 6G Roadmap

In the US and Canadian geography, the organizational effort falls on the "Next G Alliance," originating as a private sector-led initiative under the umbrella of the Alliance for Telecommunications Industry Solutions (ATIS).[28] The flagship publication "North American 6G Roadmap Priorities," finalized by the National 6G Roadmap Working Group (RWG) and consolidated in updates such as May 2024, crystallizes the region's expectations to standardize agile systems against international adversaries and local demands.[12]

The implementation horizon outlined by the Next G Alliance identifies that technologies must reach clear thresholds throughout the decade: an initial phase where early AI applications are integrated limitedly towards 2025, progressing systematically towards joint advanced optimizations via AI and full ratification oriented towards the commercial time horizon of 2030 to 2032, ensuring compliance with strict sub-millisecond Key Performance Indicators (KPIs).[9]

The alliance's policy is firmly articulated in conquering what they define as six "Audacious Goals," which simultaneously serve as structural remedies to the bottlenecks present in North American 5G infrastructures [9]:

1. **Trust, Security, and Structural Resilience:** Given the prevalence of cyber threats backed by adversary state actors and the rapid approach of the quantum supremacy day (where modern encryption will collapse), 6G must establish baseline post-quantum cryptographic methodologies in its air protocols, preventing critical network outages during crisis scenarios.[12]

2. **Immersive Experience and Enhanced Digital World:** Transcending the touchscreen era by fostering rich multi-sensory interactions that integrate haptics and volumetric feedback in human-machine and machine-machine collaborations.[31]

3. **Cost-Efficient Solutions and Information Intersections:** Vast rural geographic extensions in North America expose that the economic equations of 5G deployment are financially ruinous given low user densities. 6G must dramatically reduce structural equipment costs and create cost-efficient relay networks on a large scale.[9]

4. **Distributed Cloud and Communication Systems:** Breaking down geographic restrictions by natively and seamlessly uniting cell connectivity with heavily distributed cloud computing operations, pushing virtualization to the extreme edges of the network perimeter.[9]

5. **AI-Native Wireless Networks:** Completely transforming the protocol base to allow operational flows where machine learning algorithms continuously control physical antenna vectors without closed deterministic human programming routines.[9]

6. **Sustainability and Environmental Remediation:** Integrating solid mitigation principles and the rigorous adoption of circular economy laws aimed at reducing direct emissions from massive networks and the terminal equipment itself throughout the entire useful cycle.[9]

Additionally, the document prioritizes the imperative exploitation of mid-range spectral bands (7–15 GHz and up to 24 GHz), seeking early and massive connectivity provision versus the complex development of hardware purely focused on millimeter waves (mmWave).[12] A crucial North American requirement is the construction of a radically versatile platform; unlike 5G-Advanced (inclined towards Enhanced Mobile Broadband or Fixed Wireless Access), 6G must deeply integrate exclusive and immediate needs of disparate domains such as the Industrial Internet of Things (IIoT), defense tactical uses, and public safety citizen initiatives through deep customizable compartments.[12]

Europe: The Hexa-X Consortium, Deliverables, and Fundamental Societal Values

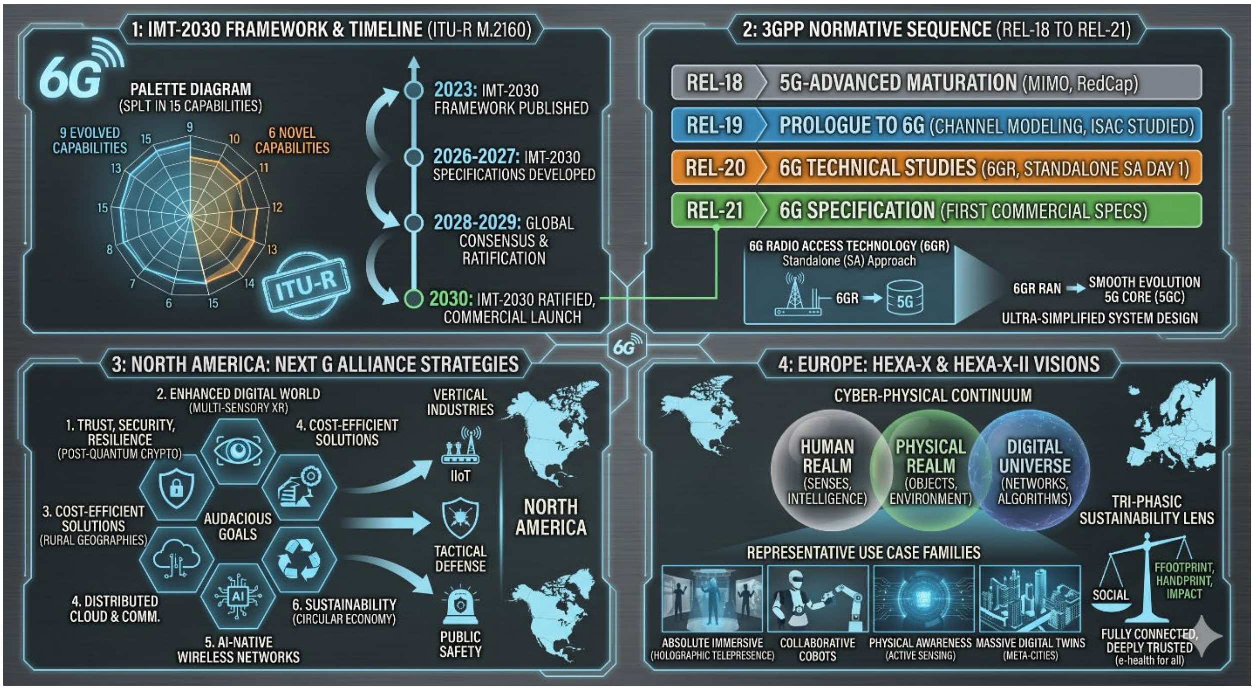
In parallel, the European sociotechnical design is channeled almost entirely by the monumental flagship consortia called Hexa-X and its iterative successor Hexa-X-II, heavily sponsored by the European Union's Horizon 2020 framework programs and operating under collaborations of massive ecosystems like Ericsson, Nokia, and leading continental universities.[10] The European geoeconomic approach differs from North America by placing an immensely central weight on intrinsic social values of inclusion and sustainability as a non-negotiable prerequisite for technological validation, something deeply codified in public deliverables such as document "D1.2 - Expanded 6G vision, use cases and societal values".[10]

The central narrative of the 6G vision proposed by Hexa-X enunciates the forced creation of an unbreakable "cyber-physical continuum".[1] In this ideological and architectural conceptualization, the three disjoint realms (the organic human world including its senses, intelligence, and anatomy; the physical ecosystem of material objects and climate environments; and the digital universe of communication networks and algorithmic calculation) will be symbiotically unified through ubiquitous and omnipresent sensory network capabilities.[1]

Unlike past cycles, Hexa-X imposes strict methodologies to frame research through what it defines as primary representative use case families, which justify technical gaps and challenges such as global coverage or the interconnection of massively distributed algorithmic AI across the telecommunications fabric [1]:

- **Absolute Immersive Experience:** Integration that surpasses conventional mobile phones and current XR interfaces, underpinning the interactive, remote, and fluid creation of holographic telepresence platforms, multi-modal educational environments, and shared social dynamics indistinguishable from physical bodily encounters through high-definition millisecond tactile feedback.[10]
- **From Robots to Collaborative Cobots:** The evolution and interoperability of embodied automaton systems to transition from encapsulated platforms (programmed robots) to autonomous collaborative mobile agents (cobots) that utilize ultra-reliable cyber-secure connections and ultra-fast edge processing (MEC), interacting dynamically and physically in the proximity of human operators in vast and fluid modern manufacturing plants.[1]
- **Physical Awareness through Active Environmental Sensing:** Cases utilizing a network as a massive geographic sensor, offering network-assisted mobility at a metropolitan level (e.g., orchestrated and preventive control multiple nautical miles around intelligent vehicles maneuvering in low visibility conditions or congested streets), a concept totally dependent on the native ISAC (Integrated Sensing and Communication) of 6G.[10]
- **Massive Digital Twins:** A framework to capture and feedback continuous, three-dimensional, and volumetric data in real-time about extensive industrial systems or mega-cities, to enable virtual replicas that allow preventive optimization of the natural world, predictive monitoring, or the immediate alteration of the underlying command chain.[1]
- **Fully Connected and Deeply Trusted Environments:** Drastically reinforcing universal global inclusion to democratize health ("e-health for all"), access to knowledge, educational equity, and in parallel implementing a pervasive ecosystem of inviolable privacy where shared data, even between heterogeneously hyperconnected nodes, is shielded from the foundational layer against espionage without the friction of bulky additive interfaces.[1]

This detailed sociological framework dictates that any architectural deployment must be validated through a strict tri-phasic sustainability lens that exceeds the pure conservation of base station megawatt consumption.[10] It must holistically contemplate the social aspects (real and tangible promotion of reducing digital disparities by providing ultra-cheap connectivity to vast underserved territories via uninterrupted satellite integration); the environmental domains (managing the handprint and footprint impact analyzed in Section 1); and, very characteristically of the Hexa-X-II consortium, sustainable micro and macroeconomic impacts, ensuring logical incentives that validate enduring productive value chains across multiple business ecosystems for the various actors involved in a complex 6G "network of networks" fabric.[1]



Block 3: Second and Third-Order Implications in the Ecosystem

The conjunction of the resolutions aimed at **mitigating the factors leading to the chronic exhaustion of 5G-Advanced** base structures with the ironclad chronological rigidity of 3GPP engenders incalculable systemic consequences across the global information technology spectrum.

3.1. The Era of Severe Functional Disaggregation

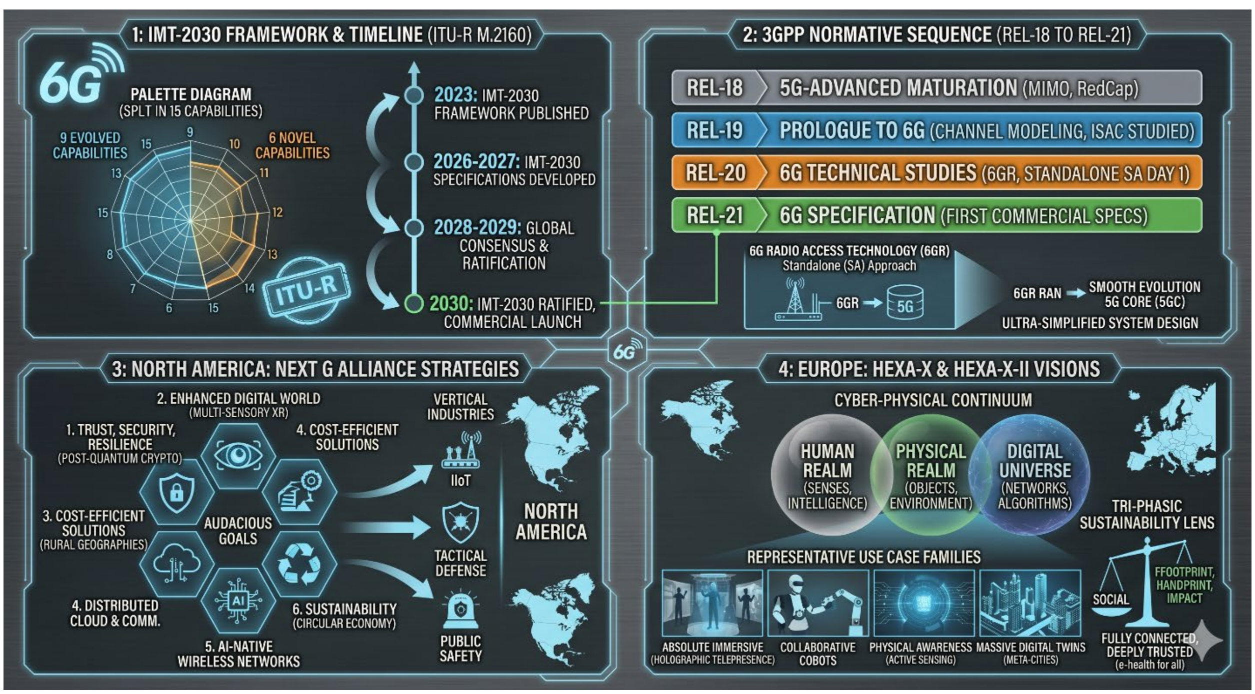
Given that the 6G landscape inherently requires instituting and solidifying a vastly distributed "cyber continuum" to the outer edge [9], conventional hegemonic architectures that centralize functionalities in switches and closed physical arrays based on unified and inscrutable hardware chassis from dominant providers are approaching a definitive dissolution. Consequently, as inferred through binding studies [1], the **transition towards cloud-native solutions will force telecommunications operators to transmute from being fixed byte pipe carriers** into cognitive providers, dynamic hybrid orchestrators that intermediate on-demand stochastic latency, processing capabilities of interconnected edge nodes, and interim swarms based on pure algorithmic inference driven by **AI and Machine Learning** across an entire regionally disseminated fractal topology.

3.2. Quantum Threats and the Imperative Integrated Defense

As the **expansive contact footprint** massively brought about by trillions of interconnected passive sensors, Agentic AI, and the massive fabric of digital twins asymptotically **multiplies the attack surface** of the super network [3], classical cybernetic axioms of "overlay protection" or post-factum complementary protocols (add-ons) **will lack structural viability.**[1] Even more alarming, and an aspect highlighted by the defensive and socio-political strategic framework articulated by the "Next G Alliance" Audacious Goals, the mature phase and **implementation of 6G will synchronously coincide with the widespread ascent of the technological level of operative quantum computing on a global scale.**[12] Therefore, current **asymmetric public key cryptographic methodologies,** current guarantors of the centralized 5G protocol, will yield to brute force, imposing a total requirement for the early developers of 3GPP Release 21 to **incorporate foundational post-quantum cryptography and deeply linked algorithms** without trustworthiness fissures across the entire physical layer (PHY) substrate of radio transmission airframes.[1]

3.3. Mitigation of Financial Value Destruction through Structured Spectral Sharing

Observing the corporate pain suffered early in the 5G deployment cycle, where schizophrenic variations of forced Non-Standalone (NSA) initial deployments severely divided and annulled rapid interoperability for vertical use cases and paralyzed the microchip market [7], **the industry demands extreme safeguards.** Thus, a profound strategic structural discovery for the baseline 6G specification is the immense priority that studies will give to efficient mechanisms known as Multi-RAT Spectrum Sharing (MRSS).[7] Early iterations of 4G and its initial coexistence with 5G were operationally cumbersome due to the heavy permanent referential flows required (overhead). In stark contrast, **the 5G air scheme** (much leaner in idle noises) **will allow interlacing and hybridizing clean transmissions and concurrent 6G overlays** over the same physical spectrum for an operationally negligible punitive margin percentage-wise.[7] This implication will ensure that **the early basic territorial propagation of 6G emerges almost instantaneously superimposed on the current 5G grid by massive global radio-telecommunication operators**, granting profitability and direct economic return capacity while the arduous physical capillary establishment of the micro cells forcibly required in millimeter waves (mmWave) and sub-THz progresses gradually and physically.[7]



Digital Twins

Block 4: Physical Layer Innovations & Enabling Hardware

The physical layer (PHY) of 6G is moving beyond the constraints of discrete antenna arrays toward continuous apertures and reconfigurable environments. This shift is necessitated by the requirement to operate in the sub-Terahertz (sub-THz) and Terahertz (THz) bands, where propagation challenges such as extreme path loss and atmospheric absorption peaks demand novel hardware solutions.

4.1. RIS & Holographic MIMO: Metasurface Control and Sub-THz Beam-Steering

Reconfigurable Intelligent Surfaces (RIS) and Holographic MIMO (HMIMO) represent the evolution of spatial dimensions in wireless communications. Traditional MIMO systems rely on discrete antenna elements, which become increasingly complex and power intensive as the frequency rises. In contrast, RIS technology utilizes arrays of passive or active scattering elements—typically meta-atoms based on varactor diodes, PIN diodes, or liquid crystals—to control the phase, amplitude, and polarization of incident waves without requiring additional radio frequency (RF) chains.

The physics of RIS is grounded in the generalized Snell's law, where an abrupt phase shift $\Phi(x,y)$ introduced at the interface allows for the manipulation of the reflected or transmitted wavefront. For an incident wave with a wavevector k_i, the direction of the reflected wave k_r is determined by:

$$\sin(\theta_r) - \sin(\theta_i) = (1/k_0) * (d\Phi/dx)$$

This capability **allows the RIS to act as a "smart mirror"** that can **redirect signals around obstacles**, effectively **maintaining Line-of-Sight (LoS) or creating favorable Non-Line-of-Sight (NLoS) paths** in dense urban environments. In the sub-THz regime (0.1–0.3 THz), where signals are highly directional and prone to blockage by even small objects, RIS is critical for coverage enhancement.

Holographic MIMO extends this concept by increasing the density of elements until the aperture appears continuously relative to the wavelength. This transition enables spatial wave field synthesis and spatial spectral holography, allowing the system to focus energy not just in specific directions (beamforming) but at specific points in space (beam-focusing). This is particularly relevant in the near-field (Fresnel) region, which expands significantly at THz frequencies. The near-field distance d_NF is defined by the Rayleigh distance:

$$d\_NF = 2D^2 / \lambda$$

where D is the largest dimension of the antenna aperture and λ is the wavelength. For a 10 cm aperture at 300 GHz, the near-field region extends up to 20 meters, allowing HMIMO to exploit the extra degrees of freedom provided by the curvature of the wavefront.

Technology Aspect	**Conventional Massive MIMO**	**RIS-Assisted MIMO**	**Holographic MIMO**
Element Spacing	λ/2	Variable (Sub-wavelength)	≪ λ/2 (Continuous)
Power Consumption	High (Active RF chains)	Very Low (Passive reflection)	Moderate (Dense integration)
Wavefront Control	Discrete Beamforming	Phase-Shift Reflector	Spatial Spectral Holography
Frequency Target	Sub-6 GHz, mmWave	mmWave, Sub-THz	Sub-THz, THz

4.2. Photonic Transceivers & THz Electronics: Overcoming the "THz Gap"

The "**THz gap**" refers to the spectral region between 0.1 THz and 10 THz, which has historically been **difficult to access** because it lies above the efficient operating range of traditional transistor-based electronics and below the range of optical lasers. 6G research aims to bridge this gap through a **combination of high-frequency semiconductor technologies and microwave photonics.**

In the electronic domain, Indium Phosphide (InP) and Gallium Nitride (GaN) are the primary semiconductor materials driving THz transceiver development. InP High Electron Mobility Transistors (HEMTs) and Heterojunction Bipolar Transistors (HBTs) offer high cut-off frequencies (f_t) and maximum oscillation frequencies (f_{max}) exceeding 1.5 THz, making them ideal for low-noise amplification and signal generation in the D-band (110–170 GHz) and beyond. GaN, conversely, is preferred for high-power amplifiers (HPAs) due to its wide bandgap and high breakdown voltage, which are essential for generating sufficient EIRP (Equivalent Isotropically Radiated Power) to overcome the high path loss at THz frequencies.

Microwave photonics offers an alternative approach by performing signal distribution and processing in the optical domain before converting it to THz radiation. For example, the TERA6G research project utilizes Silicon Nitride (Si_3N_4) waveguides and TriPleX technology to create scalable optical beamforming networks (OBFNs). By mixing two laser signals with a frequency difference in the THz range on a high-speed photodetector (e.g., a Uni-Traveling-Carrier Photodiode or UTC-PD), ultra-wideband THz signals can be generated with high stability and low phase noise.

Experimental testbeds have already demonstrated the potential of these hardware architectures. For instance, a research testbed demonstrating simultaneous 2x2 MIMO at 142 GHz and 285 GHz achieved a total data throughput of 126.915 Gbps using a 16 QAM modulation scheme over a 12.5 GHz bandwidth. The physics of such systems involves mitigating LO (Local Oscillator) feedthrough and undesired sidebands using waveguide bandpass and highpass filters, as well as addressing time alignment errors (TAE) that arise from physical differences in the converter paths and antenna structures.

4.3. AI-Native Air Interface: Autoencoder PHY and Semantic Communication

6G defines a fundamental **shift from "AI-assisted" to "AI-native" networks**. In the air interface, this translates to the use of **deep learning (DL)** to replace or optimize traditional signal processing blocks. 3GPP Release 20 and 21 research identifies three developmental phases for the AI-Native Air Interface (AI-AI).

Phase 1: Component Replacement. ML models are used to enhance or replace specific blocks, such as channel estimation, symbol demapping, or physical random-access channel (PRACH) detection.

Phase 2: Joint Optimization. ML models take on the joint role of multiple processing blocks, such as joint channel estimation, equalization, and demapping. This phase targets higher gains by allowing the system to learn bespoke waveforms adapted to specific hardware limitations, such as non-linear power amplifiers or low-resolution quantizers.

Phase 3: AI-Designed Air Interface. In this paradigm shift, AI is given the freedom to design parts of the PHY and MAC layers itself. Standardization shifts from specifying fixed modulation schemes (e.g., 64 QAM) to specifying the procedures for distributed end-to-end training and optimization.

A critical component of this AI-native design is Semantic Communication. Traditional communications are governed by Shannon's classical theory, which focuses on the reliable transmission of bits (the technical level). Semantic communication, however, operates at the "meaning" and "effectiveness" levels. **It uses a**

Semantic Knowledge Base (SKB) and Information Bottleneck (IB) theory to compress data based on its relevance to the specific task.

The Information Bottleneck principle aims to find a compressed representation Z of the input X that preserves the maximum information about the relevant task variable Y, minimizing the mutual information I(X;Z) while maximizing I(Z;Y). This allows the 6G system to transmit only the "semantic essence" of the information, potentially reducing the required bandwidth by orders of magnitude for applications like remote robot control or collaborative sensing.

Block 5: Network Architecture Evolution

The 6G architecture is evolving from a rigid cellular topology toward a user-centric, distributed, and fluid network fabric. This evolution is characterized by the implementation of Cell-Free Massive MIMO, the integration of quantum-safe security, and the fusion of computing and networking into a seamless "computational continuum".

5.1. Cell-Free & Distributed Massive MIMO: Transition to User-Centricity

Cell-Free Massive MIMO (CF-MMIMO) represents the ultimate evolution of the Distributed Antenna System (DAS). In CF-MMIMO, the traditional concept of a "cell" with defined boundaries is eliminated. Instead, a large number of geographically distributed access points (APs) are connected to a Central Processing Unit (CPU) and jointly serve all users in the same time-frequency resource.

The primary advantage of CF-MMIMO is the mitigation of inter-cell interference and the elimination of cell-edge effects, providing a nearly uniform experience to all users regardless of their location. The signal received at the n-th AP in the uplink can be expressed as:

$$y_{UL,n} = \Sigma(k=1 \text{ to } K) [g_{k,n} * s_k] + z_n$$

where $g_{k,n}$ represents the channel from the k-th user to the n-th AP, s_k is the transmitted signal, and z_n is the additive white Gaussian noise.

However, the practical implementation of CF-MMIMO faces significant engineering challenges:

Fronthaul Synchronization: Coherent joint transmission and reception require extremely precise time and frequency synchronization across all distributed APs. Any phase misalignment can lead to constructive or destructive interference, degrading the beamforming gain.

Fronthaul Capacity: The sheer volume of raw IQ data that must be exchanged between the APs and the CPU can easily saturate existing fiber or wireless backhaul links, necessitating advanced data compression techniques and the move to THz fronthaul.

Channel Acquisition: In Time Division Duplexing (TDD) mode, channel reciprocity is used to estimate downlink channels from uplink pilots. However, hardware impairments (HWIs) in the RF chains of the distributed APs can compromise this reciprocity, requiring frequent calibration.

5.2. Quantum Security: QKD and PQC Integration Framework

The advent of quantum computing poses a significant threat to the current cryptographic landscape. Algorithms such as Shor’s algorithm can efficiently factor large integers and solve discrete logarithm problems, rendering public-key schemes like RSA and Elliptic Curve Cryptography (ECC) vulnerable to decryption. 6G addresses this through a dual-pillar quantum-safe framework:

Post-Quantum Cryptography (PQC): These are software-based cryptographic algorithms designed to be secure against both quantum and classical computers. 3GPP research is focusing on integrating lattice-based, code-based, or hash-based algorithms (e.g., CRYSTALS-Kyber for key exchange and CRYSTALS-Dilithium for digital signatures) into the existing security architecture. A key challenge is the increased key size and computational overhead, which must be optimized for resource-constrained 6G devices.

Quantum Key Distribution (QKD): QKD utilizes the principles of quantum mechanics (e.g., the Heisenberg Uncertainty Principle and the No-Cloning Theorem) to exchange symmetric encryption keys with information-theoretic security. Any attempt by an eavesdropper to intercept the quantum state is instantly detected.

Within the 6G framework, QKD is envisioned for securing the backbone and fronthaul communications between core network nodes and distributed APs. Furthermore, Satellite-based QKD (SatQKD) is being explored to enable long-distance quantum key exchange across the Non-Terrestrial Network (NTN). Experimental results of quantum-encrypted NTN integration have demonstrated a Round-Trip Time (RTT) overhead of only 5 ms (approx. 0.75%), indicating that security enhancement is technically feasible for next-generation networks.

5.3. Quantum Homomorphic Encryption (QHE): Privacy-Preserving Computation at the Edge

Quantum Homomorphic Encryption (QHE) empowers 6G edge computing nodes to process information without executing prior decryption routines. QHE applies sequences of quantum logic gates directly onto encrypted information states. This establishes a Zero-Trust network architecture, allowing third-party servers to execute machine learning inference workloads or process massive biometric telemetry flows without ever accessing the semantic state of the user's original data.

The fundamental mathematical imperative requires that processing a function f on an encrypted data point $E(x)$ results in an output identical to the encryption of the function evaluated on the original data: $f(E(x)) = E(f(x))$. Classical Fully Homomorphic Encryption (FHE) architectures generate data expansion and processing overhead that are unviable for telecommunications. QHE transfers the computational load to distributed quantum clusters and employs lattice-based schemes, simultaneously ensuring cryptographic immunity against brute-force attacks executed by adversarial quantum hardware.

The technical integration of QHE into the 6G physical layer faces a critical temporal barrier. The 6G standard demands closed-loop latencies strictly below 1 ms to enable cyber-physical systems and autonomous robotics. The execution speed of quantum logic gates and qubit coherence times impose temporal bottlenecks that currently exceed this operational limit. Standardization in future 3GPP Releases requires aligning the switching speeds of edge quantum servers with the transfer rates of sub-THz links to prevent the collapse of key performance indicators.

5.4. Computational Continuum: The "Osmotic" Fusion of Cloud, Edge, and Far-Edge

6G networks will not just be about connectivity; they will be a distributed computing platform. The "Computational Continuum" refers to the seamless integration of resources across the Cloud, Edge, and Far-Edge (client devices). This architecture is governed by the principles of "Osmotic Computing," where microservices encapsulated in lightweight containers (e.g., Docker, WebAssembly) are opportunistically deployed and migrated across infrastructure nodes based on real-time load, context, and QoS requirements.
The hierarchy of the computational continuum consists of:
Far-Edge/IoT Layer: Localized computing on end devices and sensors to minimize data transmission and maximize privacy.
Edge Layer: Gateways and access points capable of performing inference and low-latency processing closer to the user.
Cloud Layer: Centralized data centers for high-performance training of AI models and long-term data analytics.
Native AI orchestration is the "digital brain" of this continuum. It manages the feedback loop between infrastructure performance and application needs, ensuring that model training and prediction tasks are performed at the most efficient location. For instance, a 6G system could prioritize Edge AI for low-latency robotic control while offloading complex city-scale simulation tasks to the Cloud.

Block 6: Transformative Use Cases

The technological pillars of 6G enable a new generation of use cases that were previously hindered by the latency and bandwidth limitations of 5G. These use cases are categorized into six scenarios under the IMT-2030 vision.

6.1. Holographic Telepresence & Internet of Senses

Holographic telepresence aims to provide lifelike 3D representations of remote individuals, requiring the transmission of multi-view video or light-field data. The "Internet of Senses" expands this by incorporating haptic feedback, allowing users to "touch" digital objects.

Technical Requirements:

Bandwidth: Transmitting high-resolution, 360-degree 3D holograms requires data rates from 1 Gbps to 1 Tbps.

Latency: Real-time haptic feedback requires an end-to-end delay of less than 1 ms to prevent sensory dissonance.

Update Rate: Haptic applications require a synchronization update rate of 1 kHz (every 1 ms), and 6G envisions up to 10 kHz for extreme precision.

6.2. Massive Real-Time Digital Twins (City-Scale)

Digital Twins (DTs) are virtual, high-fidelity replicas of physical systems. 6G will enable "Massive Twinning" of entire cities, allowing for real-time simulation, predictive maintenance, and emergency response optimization. This use case relies heavily on the integration of sensing and communication (ISAC).

Technical Requirements:

Synchronization: Real-time synchronization between the physical and digital twins requires over-the-air delays below 0.1 ms and synchronization accuracy better than 100 ns.

Positioning: Centimeter-level positioning accuracy is required to map mobile assets within the digital replica.

Data Throughput: 6G-enabled DT frameworks have demonstrated 15.6x improvement over Wi-Fi 6, achieving sub-1ms end-to-end latency for industrial bearing fault detection.

6.3. UAV/CAV Swarms and Collaborative Robotics

Collaborative swarms of Unmanned Aerial Vehicles (UAVs) and Connected Autonomous Vehicles (CAVs) require ultra-reliable low-latency communication (URLLC) to maintain coherent formations and coordinated sensing.

Technical Requirements:

Connectivity Density: 6G must support over 100 devices per cubic meter to enable dense swarms.

Reliability: Success probability of transmission must exceed 99.999%.

Jitter: Deterministic timing with microsecond-level jitter is essential for maintaining formation stability in dynamic environments.

6.4. Zero-Energy IoT (RF Energy Harvesting)

"Zero-Energy" devices are ultra-low-cost sensors that operate without batteries by harvesting energy from ambient RF signals (e.g., from base stations or Wi-Fi) or through dedicated wireless power transfer (WPT).

Technical Metrics:

Power Consumption: These devices typically consume microwatts to milliwatts and rely on passive backscattering rather than active transmission.

Bit Rate: When activated, backscattering can support burst rates of up to 40 Mbps in short 3.5 ms packages.

Energy Harvesting Coverage: The operational range depends on the harvestable power density and the measurement rate. 6G networks must support the massive, sporadic access patterns of these trillions of devices.

6.5. Brain-Computer Interfaces (BCI) and Advanced E-Health

BCIs enable direct bidirectional communication between the central nervous system (CNS) and external devices. 6G is the first generation capable of supporting the massive data flow required for high-fidelity neural streaming.

Technical Requirements:

Reliability: Mission-critical e-health services like telesurgery require a reliability of $1 - 10^{-9}$.

Latency: End-to-end latency must be less than 0.1 ms for closed-loop neural feedback.

Data Rate: Advanced medical imaging and neural sensing require data rates between 100 Gbps and 1000 Gbps.

Use Case	Data Rate	Latency	Reliability	Specific KPI
Holographic Presence	100 Gbps - 1 Tbps	< 1 ms	99.99%	Jitter < 10 µs
Digital Twin (City)	10 - 100 Gbps	< 0.1 ms	99.9999%	Sync < 100 ns
UAV / CAV Swarm	100 Mbps - 1 Gbps	1 µs - 1 ms	99.999%	Density $100/m^3$
Zero-Energy IoT	100 bps - 40 Mbps	N/A	Variable	Harvesting µW
BCI / Telesurgery	100 Gbps - 1 Tbps	0.1 ms	99.99999%	$1 - 10^{-9}$ loss

6.6. High-Fidelity Digital Twins and Massive Industrial Control

The implementation of Digital Twins in the 6G ecosystem represents the absolute convergence between the physical world and its virtual representation. Unlike the limited capabilities of 5G, the 6G standard enables the massive, real-time synchronization of high-fidelity digital replicas for critical infrastructure, smart cities, and complex biological systems. This parity demands peak data rates in the Terabit-per-second range and air interface latencies strictly below 0.1 ms to ensure that the digital model's state always remains identical to the physical object.

Integrated Sensing and Communication (ISAC) serves as the primary enabler in this scenario. It allows the network infrastructure to function as a distributed high-resolution radar, mapping the physical environment and updating the digital twin without relying exclusively on external sensors. This facilitates predictive orchestration and high-precision dynamic simulations, optimizing industrial processes, autonomous traffic flow, and smart building energy management through proactive scenario analysis before physical execution.

This paradigm is supported by the "computational continuum," where the processing load required to maintain twin fidelity is dynamically shifted between the terminal device, the network edge, and the centralized cloud. Furthermore, the integration of advanced security protocols ensures that these replicas—which contain sensitive intellectual property or biometric data—can be processed and analyzed on third-party servers without risking the exposure of the original information.

Block 7: Conclusions

The transition to 6G is an engineering endeavor of unprecedented scale, requiring the synchronization of hardware, software, and international policy.

7.1. Technical Viability: IMT-2030 KPIs vs. Hardware Limitations

While the target KPIs for IMT-2030 represent a ten-to-twenty-fold improvement over 5G, the technical viability is currently constrained by several hardware bottlenecks. For instance, the efficiency of Power Amplifiers (PAs) at THz frequencies remains significantly lower than 40% typical of 5G products. Furthermore, the beam-splitting effect in ultra-massive MIMO systems at THz frequencies complicates beam management, as signals at different subcarriers may point in different spatial directions. Additionally, the high-power consumption of dense ADCs (Analog-to-Digital Converters) in high-bandwidth transceivers necessitates the exploration of low-resolution or one-bit quantizers, which in turn require complex signal processing to mitigate quantization noise.

7.2. Critical Barriers: Spectrum, Geopolitics, and Sustainability

The stabilization of the 6G ecosystem faces three critical barriers:

Spectrum Availability: The industry estimates a requirement for an additional 2–3 GHz of mid-band spectrum (specifically in the 4.4–15.35 GHz range) to support the wide contiguous channels (400–750 MHz) needed for 6G performance targets. Decisions at the World Radiocommunication Conference 2027 (WRC-27) will be decisive for international harmonization.

Geopolitical Fragmentation: Diverse regional approaches to spectrum allocation—such as the U.S. focus on the 7.125–7.4 GHz band versus the European RSPG's long-term vision for the upper 6 GHz band—could lead to a fragmented global market, hindering roaming and increasing device costs.

Energy Sustainability: 6G must achieve a massive "handprint" by improving energy efficiency in other industries while minimizing its own "footprint". The target is for 6G to be 10x to 100x more energy-efficient per bit than 5G, ensuring that total network power consumption does not spiral out of control as traffic increases.

7.3. Final Horizon: The 2030–2035 Roadmap

The global 6G stabilization roadmap follows a structured progression:

2024–2026 (Requirement Phase): ITU-R defines technical performance requirements and evaluation methodologies. 3GPP Release 20 studies focus on use cases and service requirements.

2027–2028 (Normative Phase): 3GPP Release 21 will develop the first set of 6G technical specifications, leading to a "code freeze" by late 2028 or early 2029.

2029–2030 (Deployment Phase): Initial commercial 6G systems are expected to be available, coinciding with the official ITU designation of IMT-2030 technologies.

2030–2035 (Maturation Phase): Widespread adoption of holographic services, massive real-time digital twins, and the integration of AI-native orchestration across the global computational continuum will mark the stabilization of the 6G era.

Ultimately, 6G is more than an incremental improvement in speed; it is the fundamental infrastructure for a world where intelligence is pervasive, and connectivity is as ubiquitous as the air we breathe. The successful realization of this vision depends on the industry's ability to bridge the THz gap, master the AI-native air interface, and establish a secure, sustainable, and globally harmonized architectural foundation.

Bibliography

[1] Keysight Technologies. "6G Sub Terahertz MIMO." keysight.com.

[2] MDPI. "6G Networks and the AI Revolution—Exploring Technologies, Applications, and Emerging Challenges." mdpi.com.

[3] Huawei. "6G: The Next Horizon." www-file.huawei.com.

[4] ITU. "IMT-2030: Technical requirements for the 6G future." itu.int.

[5] IEEE ComSoc. "IMT-2030 ("6G") Minimum Technology Performance Requirements." techblog.comsoc.org.

[6] NGMN. "ITU-R framework for IMT-2030: review and future direction." ngmn.org.

[7] WirelessBrew. "Comprehensive Catalog of 6G Study Items (Rel-20)." wirelessbrew.com.

[8] one6G Summit 2025. "6G Roadmap & Status in 3GPP RAN." summit2025.one6g.org.

[9] Ericsson. "6G standardization timeline and principles." ericsson.com.

[10] IEEE Xplore. "RIS-Assisted Terahertz Communications for 6G." ieeexplore.ieee.org.

[11] ResearchGate. "White paper: Fundamentals of THz technology for 6G." researchgate.net.

[12] SMI Doctoral School (ED432). "Ph.D. thesis proposal 2022 call Title: Deep learning techniques for RIS-aided 6G communications in mmWaves and sub-THz bands." cedric.cnam.fr.

[13] University of Waterloo. "RIS-Aided Cell-Free Massive MIMO Systems for 6G: Fundamentals, System Design, and Applications." uwaterloo.ca.

[14] MDPI. "Key Enabling Technologies for 6G: The Role of UAVs, Terahertz Communication, and Intelligent Reconfigurable Surfaces in Shaping the Future of Wireless Networks." mdpi.com.

[15] Wireless Lab at UCLA / arXiv. "A Survey on Advancements in THz Technology for 6G: Systems, Circuits, Antennas, and Experiments." wireless.ee.ucla.edu / arxiv.org.

[16] LioniX International. "TERA6G: 6G networks with photonic wireless transceivers." lionix-international.com.

[17] Semantic Scholar. "Microwave Photonics Empowered Integrated Sensing and Communication for 6G." semanticscholar.org.

[18] MDPI. "AI-Native PHY-Layer in 6G Orchestrated Spectrum-Aware Networks." mdpi.com.

[19] Nokia. "Toward a 6G AI-Native Air Interface." nokia.com.

[20] arXiv. "Way to Build Native AI-driven 6G Air Interface: Principles, Roadmap, and Outlook." arxiv.org.

[21] Frontiers / ResearchGate. "A comprehensive review of AI-native 6G: integrating semantic communications, reconfigurable intelligent surfaces, and edge intelligence for next-generation connectivity." frontiersin.org / researchgate.net.

[22] arXiv / ResearchGate. "Ultra-Dense Cell-Free Massive MIMO for 6G: Technical Overview and Open Questions." arxiv.org / researchgate.net.

[23] Science China. "Full-spectrum cell-free RAN for 6G systems: system design and experimental results." scis.scichina.com.

[24] Extreme Networks. "6G Wide Area Cloud – The Marriage of Intelligent Compute, Network, and AI Platform for Ubiquitous Global Connectivity Part 1." extremenetworks.com.

[25] ResearchGate / Scifiniti. "Quantum-Safe Networks for 6G: An Integrated Survey on PQC, QKD, and Satellite QKD with Future Perspectives." researchgate.net / scifiniti.com.

[26] ITU. "ITU-T Rec. Technical Report (03/2020) Security considerations for quantum key distribution network." itu.int.

[27] MDPI. "Quantum-Enhanced Security Framework for Next-Generation Space–Terrestrial Networks." mdpi.com.

[28] European Commission. "6G-Cloud." ec.europa.eu.

[29] PMC NIH. "Osmotic Cloud-Edge Intelligence for IoT-Based Cyber-Physical Systems." pmc.ncbi.nlm.nih.gov.

[30] ENTSO-e. "Cloud and Edge Computing." entsoe.eu.

[31] MDPI. "Enabling Tactile Internet via 6G: Application Characteristics, Requirements, and Design Considerations." mdpi.com.

[32] Frontiers. "Toward immersive communications in 6G." frontiersin.org.

[33] Next G Alliance. "6G Digital Twins Use Cases and Requirements." nextgalliance.org.

[34] Scribd. "6G Digital Twins: Use Cases & Insights." scribd.com.

[35] Nokia. "Digital twins in 6G networks." nokia.com.

[36] arXiv. "6G-Enabled Digital Twin Framework for Real-Time Cyber-Physical Systems: An Experimental Validation with Industrial Bearing Fault Detection." arxiv.org.

[37] arXiv. "URLLC for 6G Enabled Industry 5.0: A Taxonomy of Architectures, Cross Layer Techniques, and Time Critical Applications." arxiv.org.

[38] Queen's University Belfast. "Digital twin for 6G: taxonomy, research challenges, and the road ahead." pureadmin.qub.ac.uk.

[39] University of Southampton. "On the Road to 6G: Visions, Requirements, Key Technologies and Testbeds." eprints.soton.ac.uk.

[40] Lancashire Online Knowledge. "6G Vision in Developing Swarms of Collaborative Robotics." knowledge.lancashire.ac.uk.

[41] MDPI. "Measurement-Informed Latency Limits for Real-Time UAV Swarm Coordination." mdpi.com.

[42] arXiv. "A Comprehensive Survey of 5G URLLC and Challenges in the 6G Era." arxiv.org.

[43] DiVA portal. "RF Energy Harvesting for Zero- Energy Devices and Reconfigurable Intelligent Surfaces." diva-portal.org.

[44] Ericsson. "Energy harvesting in a 6G sensor-driven world." ericsson.com.

[45] Transforma Insights. "3 billion IoT devices have the potential to be powered by RF energy harvesting in 2030." transformainsights.com.

[46] Moms Across America. "BISC, 6G, and the Architecture of Human-Computer Integration." momsacrossamerica.com.

[47] IEEE Xplore. "Brain–Computer Interface—A Brain-in-the-Loop Communication System." ieeexplore.ieee.org.

[48] Tongji University. "Machine Learning for 6G Enhanced Ultra-Reliable and Low-Latency Services." aibi.tongji.edu.cn.

[49] Nokia. "Energy efficiency in next-generation mobile networks." nokia.com.

[50] Nokia. "Spectrum decisions shaping the road to 6G." nokia.com.

[51] CTIA. "Spectrum Priorities For WRC-27." ctia.org.

[52] RADIO SPECTRUM POLICY GROUP. "Interim Opinion on WRC-27." radio-spectrum-policy-group.ec.europa.eu.

[53] Nokia. "The path to 6G with unparalleled energy savings." nokia.com.

[54] AIOTI. "Energy-6G Convergence: a Perspective." aioti.eu.

[55] ADROIT6G. "6G KPIS –DEFINITIONS AND TARGET VALUES." adroit6g.eu.

[56] Chalmers University of Technology. "6G Vision, Value, Use Cases and Technologies from European 6G...". research.chalmers.se. [Accedido: 07-abr-2026].

[57] IEEE Xplore. "Towards 6G: Past, Present, and Future." ieeexplore.ieee.org. [Accedido: 07-abr-2026].

[58] IEEE Xplore. "The Interplay Between Generative AI and 5G-Advanced Toward 6G." ieeexplore.ieee.org. [Accedido: 07-abr-2026].

[59] InterDigital. "Press Releases." ir.interdigital.com. [Accedido: 07-abr-2026].

[60] ITU. "IMT-2030: Technical requirements for the 6G future." itu.int. [Accedido: 07-abr-2026].

[61] ITU. "Recommendation ITU-R M.2160-0 (11/2023) - Framework and overall objectives of the future development of IMT for 2030 and beyond." itu.int. [Accedido: 07-abr-2026].

[62] Ericsson. "6G standardization timeline and principles." ericsson.com. [Accedido: 07-abr-2026].

[63] Ericsson. "The next wave of advanced 5G – 3GPP Release 19." ericsson.com. [Accedido: 07-abr-2026].

[64] Next G Alliance. "Next G Alliance Report: Roadmap to 6G." nextgalliance.org. [Accedido: 07-abr-2026].

[65] Hexa-X-II. "Deliverable D1.2 – 6G use cases and requirements." hexa-x-ii.eu. [Accedido: 07-abr-2026].

[66] IEEE Xplore. "5G/6G Architecture Evolution for XR and Metaverse." ieeexplore.ieee.org. [Accedido: 07-abr-2026].

[67] Next G Alliance. "National 6G Roadmap Working Group: North American 6G Roadmap Priorities." nextgalliance.org. [Accedido: 07-abr-2026].

[68] Next G Alliance. "6G Library." nextgalliance.org. [Accedido: 07-abr-2026].

[69] Ericsson. "Hexa-X: 6G technology and its evolution so far." ericsson.com. [Accedido: 07-abr-2026].

[70] Qualcomm. "A closer look at 5G Advanced Release 18." qualcomm.com. [Accedido: 07-abr-2026].

[71] FirstNet.gov. "3GPP focuses on Release 20 6G and 5G-Advanced, including key public-safety needs." firstnet.gov. [Accedido: 07-abr-2026].

[72] Qualcomm. "3GPP Release 20: Completing the 5G Advanced evolution and preparing for global 6G standardization." qualcomm.com. [Accedido: 07-abr-2026].

[73] IEEE Xplore. "Mobile Networks Toward 5G/6G: Network Architecture, Opportunities and Challenges in Smart City." ieeexplore.ieee.org. [Accedido: 07-abr-2026].

[74] ITU. "M.2160: Framework and overall objectives of the future development of IMT for 2030 and beyond." itu.int. [Accedido: 07-abr-2026].

[75] UNIDIR. "IMT-2030 (6G)." unidir.org. [Accedido: 07-abr-2026].

[76] Digital Regulation Platform. "Overview of 6G (IMT-2030)." digitalregulation.org. [Accedido: 07-abr-2026].

[77] FirstNet Authority. "3GPP completes Release 19, while progress begins on 6G work." firstnet.gov. [Accedido: 07-abr-2026].

[78] 5G Americas. "Becoming 5G-Advanced: the 3GPP 2025 Roadmap." 5gamericas.org. [Accedido: 07-abr-2026].

[79] FirstNet Authority. "3GPP shifts focus toward 6G while continuing work on 5G-advanced." firstnet.gov. [Accedido: 07-abr-2026].

[80] ETSI. "3GPP 6G Roadmap." etsi.org. [Accedido: 07-abr-2026].

[81] Qualcomm. "3GPP RELEASE 20 - Completing the 5G Advanced evolution and preparing for the next-generation wireless." qualcomm.com. [Accedido: 07-abr-2026].

[82] one6G Summit 2025. "6G Roadmap & Status in 3GPP RAN." summit2025.one6g.org. [Accedido: 07-abr-2026].

[83] Next G Alliance. "Home." nextgalliance.org. [Accedido: 07-abr-2026].

[84] arXiv. "From 5G to 6G: A Survey on Security, Privacy, and Standardization Pathways." arxiv.org. [Accedido: 07-abr-2026].

[85] Sparkco. "6G Wireless Infrastructure Deployment: Strategic Industry Analysis 2025." sparkco.ai. [Accedido: 07-abr-2026].

[86] ATIS. "Overview 2023." atis.org. [Accedido: 07-abr-2026].

[87] Next G Alliance. "National 6G Roadmap." nextgalliance.org. [Accedido: 07-abr-2026].

[88] Next G Alliance. "6G Technologies - Next G Alliance Report." nextgalliance.org. [Accedido: 07-abr-2026].

[89] Next G Alliance. "AI-Native Wireless Networks." nextgalliance.org. [Accedido: 07-abr-2026].

[90] NTIA / Regulations.gov. "Before the National Telecommunications and Information Administration U.S. Department of Commerce." regulations.gov. [Accedido: 07-abr-2026].

[91] Hexa-X-II. "Hexa-X-II Project Releases Deliverable D1.2 on 6G Use Cases." hexa-x-ii.eu. [Accedido: 07-abr-2026].